Chapter 4

Modeling Causal Impact via Randomized Experiments

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4.1 Learning Treatment Effects via Randomization

Potential outcomes.

Consider an RCT with n units, indexed by $i = 1, \ldots, n$. Each unit receives a binary treatment $W_i \in \{0, 1\}$ and yields an observed outcome Y_i. For each unit, define potential outcomes $Y_i(1)$ and $Y_i(0)$, linked to the observed outcome by

$$Y_i = Y_i(W_i).$$

The individual treatment effect is

$$\Delta_i = Y_i(1) - Y_i(0).$$

The core identification challenge—often called the fundamental problem of causal inference—is that for any given unit, only one of $Y_i(1)$ and $Y_i(0)$ can be observed.

Even though Δ_i itself is unobservable, randomization allows us to learn averages of these effects. In finite samples, define the sample average treatment effect (SATE):

$$\Delta = \frac{1}{n} \sum_{i=1}^{n} \big(Y_i(1) - Y_i(0)\big).$$

If units are i.i.d. draws from a population distribution $\mathbb{P}$, define the population average treatment effect (ATE):

$$\tau = \mathbb{E}_{\mathbb{P}}[Y_i(1) - Y_i(0)]\,.$$

Difference-in-Means Estimation.

A natural estimator for measuring the causal impact of a treatment is the treated-control **d**ifference in sample means:

$$\hat{\tau}_{DM} = \frac{1}{n_1} \sum_{W_i=1} Y_i - \frac{1}{n_0} \sum_{W_i=0} Y_i, \qquad n_w = |\{i : W_i = w\}|.$$

To study this estimator, we impose two further assumptions. First, SUTVA (the Stable Unit Treatment Value Assumption):

$$Y_i = Y_i(W_i), \qquad i = 1, \ldots, n.$$

SUTVA implies two key conditions. First, it means there is no interference between units, so that unit i's outcome depends only on W_i, not on others' assignments. Second, it implies that there are no hidden versions of the treatment, meaning the intervention is uniform enough that $Y_i(1)$ represents a single, unambiguous potential outcome for each unit, rather than varying effects from different doses or versions of the treatment. Together, these rule out spillovers and allow us to cleanly interpret the observed treated-control contrast as a causal contrast between $Y_i(1)$ and $Y_i(0)$.

The second assumption is that the treatment is completely randomized conditional on a fixed number of treated units n_1:

$$\mathbb{P}(W_i = 1 \mid \{Y_i(0), Y_i(1)\}_{i=1}^n, n_1) = \frac{n_1}{n}, \qquad i = 1, \ldots, n.$$

These assumptions yield the following foundational result:

Theorem 4.1.1 (Difference in means is unbiased in a completely randomized trial)**.** *Under the assumptions above,*

$$\mathbb{E}[\hat{\tau}_{DM} \mid \{Y_i(0), Y_i(1)\}_{i=1}^n,\ n_0 > 0,\ n_1 > 0] = \Delta.$$

The proof is short and relies on linearity of expectation, so we give it here:

Proof. When $n_1 > 0$,

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{n_1}\sum_{W_i=1} Y_i\right] &= \mathbb{E}\left[\frac{1}{n_1}\sum_{i=1}^n W_i Y_i \,\middle|\, \{Y_i(0), Y_i(1)\}_{i=1}^n,\ n_1\right] \\
&= \mathbb{E}\left[\frac{1}{n_1}\sum_{i=1}^n W_i Y_i(1) \,\middle|\, \{Y_i(0), Y_i(1)\}_{i=1}^n,\ n_1\right] \quad \text{(SUTVA)} \\
&= \frac{1}{n_1}\sum_{i=1}^n Y_i(1)\, \mathbb{E}[W_i \mid \{Y_i(0), Y_i(1)\}_{i=1}^n,\ n_1] \quad (Y_i(1) \text{ is fixed in the conditioning set}) \\
&= \frac{1}{n}\sum_{i=1}^n Y_i(1). \quad \text{(Randomization)}
\end{aligned}$$

An analogous argument gives

$$\mathbb{E}\left[\frac{1}{n_0}\sum_{W_i=0} Y_i\right] = \frac{1}{n}\sum_{i=1}^n Y_i(0),$$

and subtracting the two displays yields the result. □

4.2 Recommended Reading

Historically, the potential-outcomes framework for causal inference traces its origins to Jerzy Neyman's 1923 work on agricultural experiments (See the 1990 English translation of the original, (Neyman, 1990)). Although Neyman and R.A. Fisher were developing foundational experimental methodologies during the same era (with Fisher formulating his randomization-based design methodology (Fisher, 1935)), their early efforts were independent. Both were separately motivated by the practical demands of early 20th-century agricultural research stations: Neyman at the National Institute of Agriculture in Poland (Reid, 1982), and Fisher at the Rothamsted Experimental Station in the UK (Box, 1978). Decades later, Donald Rubin modernized and extended Neyman's framework to accommodate both randomized and nonrandomized observational settings (Rubin, 1974); see also the excellent review in (Holland, 1986). Rubin is also explicitly credited with formalizing the Stable Unit Treatment Value Assumption (SUTVA) (Rubin, 1980). Theorem 4.1.1 and the framing leading up to it is based on Chapter 1 of (Wager, 2024).

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